Uniwersytet SWPS

Task list no. 12 – In this lab, we will implement the algorithm for finding a strongly connected component and the Prim algorithm to find the minimum spanning tree.

Task 1.

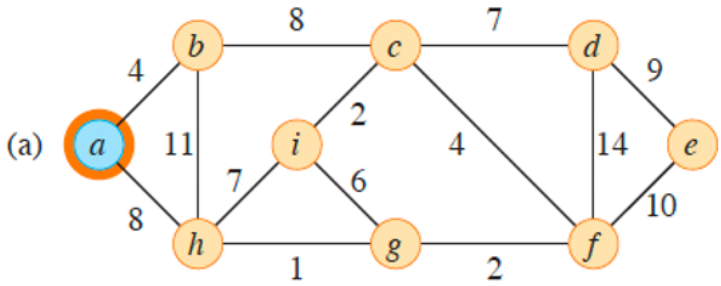
A) Using depth-first search to find the strongly connected graph for the following graph. Print the finish time of all nodes after DFS finishes, and also print all of the strongly connected components you found.

B- homework) Generate a randomly generated undirected graph with 100 nodes and at least 5000 edges, and find all of the strongly connected components in this graph.

$$A=\begin{bmatrix} 0,1,1,0,0,0,0,0,0 \\ 1,0,0,0,0,1,0,0,0 \\ 1,0,0,0,0,0,0,0,0 \\ 0,0,0,0,0,1,0,0,0 \\ 0,0,0,0,0,0,1,0,0 \\ 0,1,0,1,0,0,0,0,0 \\ 0,0,0,0,1,0,0,1,0 \\ 0,0,0,0,0,0,1,0,1 \\ 0,0,0,0,0,0,0,1,0 \end{bmatrix}$$

Task 2. Write the Prim's algorithm to find the minimum spanning tree of the following graph.

Hint: A min heap is used to implement the priority queue.



For implementation, please assign the following labels to each node of the graph as follows: a:0 b:1 c:2 d:3 e:4 f:5 g:6 h:7 g:8

A) Print the min-heap after each step of selecting a new node, and print the minimum spanning tree at the final step.

Note. The lab report must include the code and the corresponding result.

Copyright by SWPS University in Warsaw